AI in Cyber Security:

Enhancing Digital Defense

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WHAT IS ARTIFICIAL INTELLIGENCE?

AI simulates human intelligence in machines
to perform tasks like learning and decision-making.

MACHINE LEARNING
learns from data

DEEP LEARNING
uses neural networks to find complex patterns

PATTERN RECOGNITION
identifies trends and anomalies

CYBER SECURITY CHALLENGES TODAY

Cyberattacks are more **sophisticated** and harder to stop

The **volume and complexity** of threats outpace human capacity

Attackers move fast — traditional defenses **struggle to keep up**

Organizations need **real-time detection and response**

HOW AI ENHANCES CYBER SECURITY

THREAT DETECTION
AI analyzes patterns to spot attacks early

AUTOMATED RESPONSE
AI-powered actions contain threats instantly

FRAUD PREVENTION
AI models identify fraudulent activity

PREDICTIVE ANALYTICS
AI forecasts vulnerabilities before exploitation

Benefits and Risks of AI in Cyber Security

BENEFITS

Faster threat detection and response times

Ability to process vast data volumes efficiently

Continuous learning to adapt to evolving threats

RISKS

Adversarial AI attacks can manipulate systems

Overreliance on AI may reduce human oversight

Privacy concerns with data collection and analysis

THE FUTURE OF AI IN CYBER SECURITY

Explainable AI for transparent security decisions

Seamless human-AI collaboration

AI-powered threat hunting

Stronger resilience to evolving threats

“THE FUTURE OF CYBER DEFENSE IS INTELLIGENCE AUGMENTED BY AI.”

THANK YOU

FOR YOUR ATTENTION!

Questions or feedback?
Let’s continue the conversation.